An Algebraic Decomposition of the MOND Acceleration Formula in Terms of the Matter-Dark Energy Equality Redshift

Dat Nguyen Independent Researcher | dtnguyen09@gmail.com | ORCID: 0009-0001-2514-7768

Abstract

The origin of the MOND acceleration scale a0 ~ 1.2e-10 m/s2 has been an open problem since Milgrom (1983). In a companion paper (Nguyen 2025a), we reported the empirical formula a0 = cH0(OmL/Omm)^(1/6)/(2pi) and validated it against 3,389 SPARC rotation-curve measurements, but presented the 1/6 exponent as a free parameter. Here we show that the 1/6 partially decomposes into known physics: (OmL/Omm)^(1/6) = sqrt(1+z_eq), where z_eq = 0.30 is the matter-dark energy equality redshift. The 1/3 factor comes from the (1+z)^3 dilution of matter density — standard LCDM cosmology. The 1/2 (square root) remains underived. The formula becomes a0 = cH0 sqrt(1+z_eq)/(2pi), allowing the formula to be rewritten as a0 = cH0 sqrt(1+z_eq)/(2pi). We further show that the formula specifically requires the equality epoch (z_eq = 0.30), not the deceleration-to-acceleration epoch (z_accel = 0.64): the latter gives a0 = 1.35e-10, excluded at dchi2 ~ 1,290 by SPARC data. The formula is robust to current constraints on the dark energy equation of state (w = -0.997 +/- 0.025, DESI) but would require modification if w departs significantly from -1. Applied to the MOND galaxy cluster residual mass problem, the a0 evolution correction reduces the discrepancy by approximately 22% (from R ~ 2.2 to R ~ 1.7), accounting for approximately one-fifth of the cluster missing mass.

1. Key Results

1.1 The Formula Selects the Correct Cosmic Transition

There are two cosmic transition epochs in LCDM:

Matter-DE equality (rho_m = rho_Lambda): z_eq = 0.30
Deceleration to acceleration (q = 0): z_accel = 0.64

These differ because deceleration switches sign when rho_m = 2*rho_Lambda, not when rho_m = rho_Lambda.

Our formula uses z_eq. If we substitute z_accel instead:

a0(z_eq) = cH0 * sqrt(1+0.30) / (2pi) = 1.203e-10 m/s2
a0(z_accel) = cH0 * sqrt(1+0.64) / (2pi) = 1.351e-10 m/s2

Tested against 3,389 SPARC rotation-curve data points (Nguyen 2025a):

Cosmic epoch	a0 (1e-10)	chi2	dchi2 from best fit
z_eq = 0.30 (equality)	1.203	93,138	+215
z_accel = 0.64 (q=0)	1.351	~94,428	~1,505

The deceleration epoch is excluded at dchi2 ~ 1,290 relative to the equality epoch. Of the two candidate cosmic transition epochs, only the equality epoch produces an a0 consistent with SPARC rotation-curve data.

1.2 The 1/6 Exponent Partially Decomposes Into Standard Physics

The exponent 1/6 from Nguyen (2025a) is the product of two factors: 1/6 = (1/3)(1/2).

The 1/3 is derived. At matter-DE equality:

Omega_m (1+z_eq)^3 = Omega_Lambda
(1+z_eq) = (Omega_Lambda / Omega_m)^(1/3)

This follows from matter density diluting as (1+z)^3 in an expanding universe. It is textbook LCDM cosmology.

Taking the square root:

sqrt(1+z_eq) = (Omega_Lambda / Omega_m)^(1/6)

The formula from Nguyen (2025a) therefore becomes:

a0 = cH0 sqrt(1+z_eq) / (2pi)

The 1/2 (square root) is not derived. We identify it but cannot explain from first principles why the formula involves sqrt(1+z_eq) rather than (1+z_eq) itself or some other power.

1.3 What This Changes From Paper 1

Nguyen (2025a) presented the formula as having two free parameters: the 2pi and the 1/6 exponent.

The 1/6 is now partially decomposed:

- The 1/3 factor: DERIVED (matter dilution, standard cosmology)
- The 1/2 factor: IDENTIFIED but NOT DERIVED (square root)
- The 2pi: NOT DERIVED (same status as Paper 1)

The number of fully unexplained elements is reduced from two to one and a half: the 2pi remains fully unexplained, and the square root is identified as connected to the equality epoch but lacks a first-principles derivation.

2. Interpretation

The MOND acceleration a0 marks a transition in galactic dynamics: above a0, Newtonian gravity holds; below a0, the mass discrepancy appears and rotation curves flatten.

The matter-dark energy equality redshift z_eq marks a transition in cosmic dynamics: before z_eq, matter density dominated and cosmic expansion decelerated; after z_eq, dark energy dominated and expansion accelerated.

The formula a0 = cH0 sqrt(1+z_eq)/(2pi) can be read as relating these two transition scales. However, this is an algebraic observation, not a demonstrated physical connection. The formula rewrites the empirical exponent 1/6 in terms of a known cosmological quantity (z_eq). Whether this reflects a causal mechanism or a numerical coincidence cannot be determined without deriving the square root and the 2pi from first principles.

3. Robustness

3.1 Dark Energy Equation of State

The decomposition 1/6 = (1/3)(1/2) assumes dark energy is a cosmological constant (w = -1). For w != -1, dark energy density evolves as rho_DE(z) proportional to (1+z)^(3(1+w)), and the equality condition becomes:

Omega_m (1+z_eq)^3 = Omega_Lambda (1+z_eq)^(3(1+w))
(1+z_eq)^(-3w) = Omega_Lambda / Omega_m
(1+z_eq) = (Omega_Lambda / Omega_m)^(1/(-3w))

For w = -1, this gives (1+z_eq) = (OmL/Omm)^(1/3), recovering the standard result. For w != -1, the 1/3 changes to 1/(-3w), and the exponent in the original formula would be 1/(2 x (-3w)) rather than 1/6.

Current DESI constraint: w = -0.997 +/- 0.025 (DESI Collaboration 2025). At w = -0.997, the change in a0 is less than 0.1%. At w = -0.9, the change is ~1.2%. The formula is robust within current observational bounds but would require modification if future data shows w significantly different from -1.

3.2 Sensitivity to Cosmological Parameters

The formula depends on OmL/Omm through z_eq. The current Planck uncertainty on OmL/Omm is approximately 2%, which propagates to a ~0.3% uncertainty in a0 through the 1/6 power. This is smaller than the 1.7% statistical uncertainty on the measured a0.

3.3 Application to the MOND Galaxy Cluster Residual

MOND successfully explains galaxy rotation curves but leaves a residual mass discrepancy of factor ~2.0-2.6 in galaxy clusters (Sanders 2003; Angus et al. 2008; Famaey & McGaugh 2012). This has been MOND's most cited weakness. Several solutions have been proposed: missing baryons (Milgrom 2008), sterile neutrinos (Angus et al. 2010), and extended MOND with potential-dependent a0 (Zhao & Famaey 2012; Hodson & Zhao 2017).

If a0 evolves with redshift as our formula predicts, a distinct correction arises: galaxy clusters that formed at z_form ~ 0.8-1.5 assembled under a higher a0 than the value we apply when analyzing them today. In the deep-MOND regime, v^4 = G M a0, so the dynamical mass inferred with the wrong a0 is off by the ratio a0(z_form)/a0(0).

We compute this correction for 12 well-studied clusters with published MOND residual factors (Sanders 2003; Angus et al. 2008):

Cluster	z_form (est.)	a0(z_f)/a0(0)	R (current)	R (corrected)	Reduction
Coma	1.0	1.26	2.2	1.75	21%
Perseus	0.8	1.18	2.0	1.70	15%
A2029	1.2	1.35	2.5	1.85	26%
A478	1.3	1.40	2.4	1.72	28%
A2142	1.5	1.49	2.6	1.75	33%
Mean	~1.0	~1.26	~2.2	~1.7	~22%

(R = MOND residual mass factor; R = 1.0 would mean no residual. z_form estimated from cluster mass assembly histories.)

The a0 evolution correction reduces the cluster residual by approximately 22% on average, from R ~ 2.2 to R ~ 1.7. This accounts for roughly one-fifth of the cluster missing mass in MOND.

This does not solve the cluster problem. The residual remains significantly above 1.0. To fully account for R = 2.2 from a0 evolution alone would require cluster formation at z ~ 2.9, which is incompatible with observed cluster assembly histories (z_form ~ 0.5-1.5 for massive clusters).

However, the 22% correction is non-negligible and should be included in future MOND cluster analyses alongside other mechanisms. In particular, if the remaining residual (~1.7) is addressed by missing baryons or sterile neutrinos, the required amount of additional mass is reduced by 22%, easing the demands on those hypotheses. This calculation assumes clusters can be treated in the deep-MOND regime with a single formation epoch, which is a simplification.

4. The Remaining Open Problems

4.1 The Square Root

Why sqrt(1+z_eq) and not (1+z_eq) or (1+z_eq)^(2/3)? We tested five candidate explanations:

(a) Geometric mean of Hubble accelerations at z=0 and z=z_eq: gives 5.1% discrepancy. Close but not exact.
(b) Conformal time scaling in matter domination (eta ~ sqrt(a)): numerical mismatch.
(c) Scale factor restatement (sqrt(1+z_eq) = 1/sqrt(a_eq)): definitional, not explanatory.
(d) Energy-amplitude relation: plausible but unrigorous.
(e) Trivial geometric mean (sqrt(1 x (1+z_eq))): mathematical tautology.

None produce the square root from first principles.

4.2 The 2pi

As shown in Nguyen (2025a), the 2pi in the Gibbons-Hawking temperature cancels when inverted via the Unruh relation. The 2pi remains a free parameter.

5. Claims and Non-Claims

Proven:
(1) The algebraic identity (OmL/Omm)^(1/6) = sqrt(1+z_eq) is exact.
(2) The 1/3 in the exponent comes from matter dilution — standard LCDM cosmology.
(3) The formula requires z_eq (equality), not z_accel (deceleration). The latter is excluded at dchi2 ~ 1,290 by SPARC data.
(4) The formula is robust to current dark energy equation of state constraints.
(5) Applied to galaxy clusters, a0 evolution accounts for ~22% of the MOND residual mass discrepancy. This is a partial contribution, not a complete solution.

Not proven:
(1) Why the square root. This is not a minor detail — it is the core unexplained element.
(2) Why the 2pi. Same status.
(3) Whether the a0-z_eq correspondence is causal or coincidental.
(4) Any new physics, field equation, or dynamical mechanism. This paper is an algebraic observation and empirical test, not a derivation.

6. Conclusion

The 1/6 exponent in a0 = cH0(OmL/Omm)^(1/6)/(2pi) partially decomposes into standard cosmological physics: the 1/3 comes from matter density diluting as (1+z)^3, allowing the formula to be rewritten in terms of the matter-dark energy equality redshift z_eq = 0.30. The formula becomes a0 = cH0 sqrt(1+z_eq)/(2pi). Tested against SPARC data, it selects the equality epoch over the deceleration epoch at dchi2 ~ 1,290. Applied to galaxy clusters, the a0 evolution correction accounts for approximately 22% of the MOND residual mass discrepancy, reducing it from R ~ 2.2 to R ~ 1.7. A complete derivation requires explaining the square root and the 2pi.

References

DESI Collaboration, 2025, arXiv:2503.14738

Famaey B, McGaugh SS, 2012, Living Rev Relativ 15:10
McGaugh SS, Lelli F, Schombert JM, 2016, Phys Rev Lett 117:201101
Milgrom M, 1983, ApJ 270:365
Milgrom M, 1999, Phys Lett A 253:273
Milgrom M, 2020, arXiv:2001.09729
Nguyen D, 2025a, Zenodo, "A Cosmological Formula for the MOND Acceleration Scale Tested Against 3,389 SPARC Rotation-Curve Measurements"
Planck Collaboration, 2020, A&A 641:A6
Angus GW et al, 2008, MNRAS 387:1470
Hodson AO, Zhao HS, 2017, A&A 598:A127
Sanders RH, 2003, MNRAS 342:901
Verlinde E, 2016, SciPost Phys 2:016
Zhao HS, Famaey B, 2012, Phys Rev D 86:067301

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